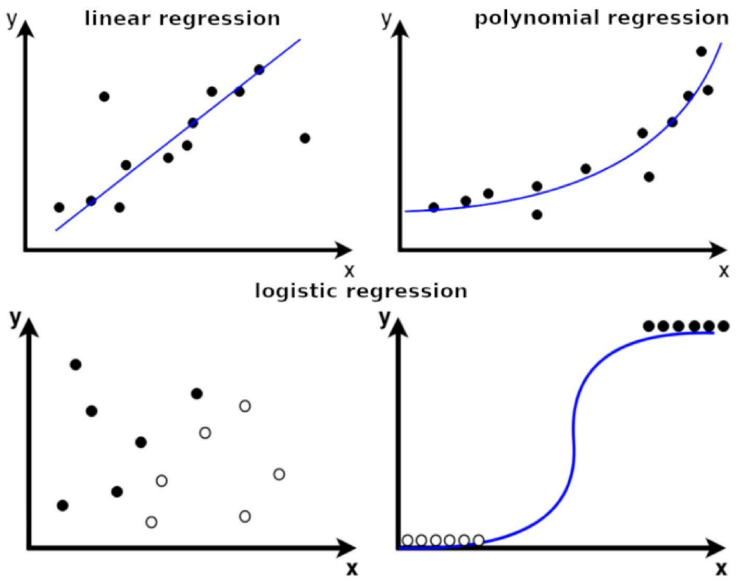
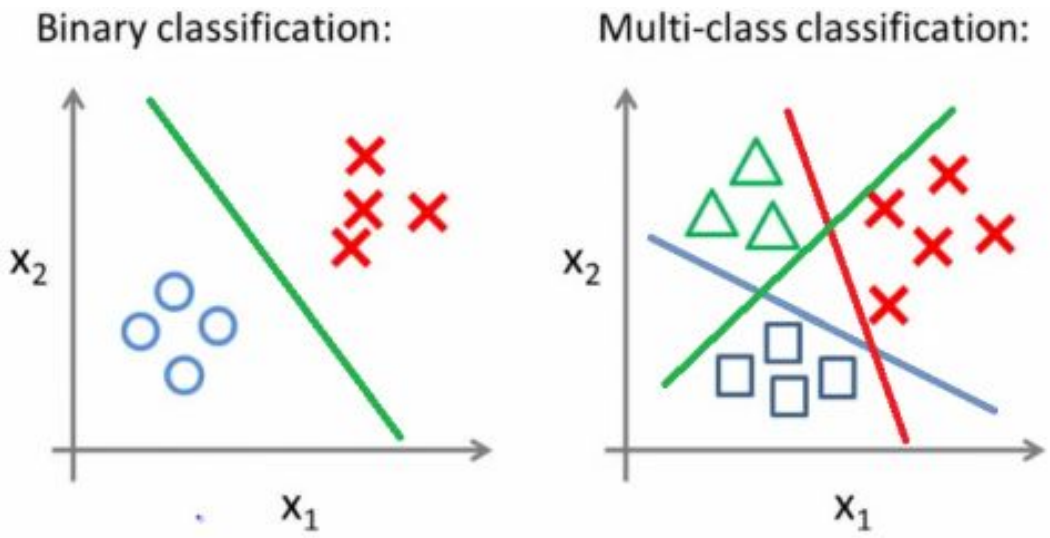
CLASSIFICATION 1: Logistic Regression Recap

DATA 201 -- WEEK 8A

SPRING 2026 UPDATE





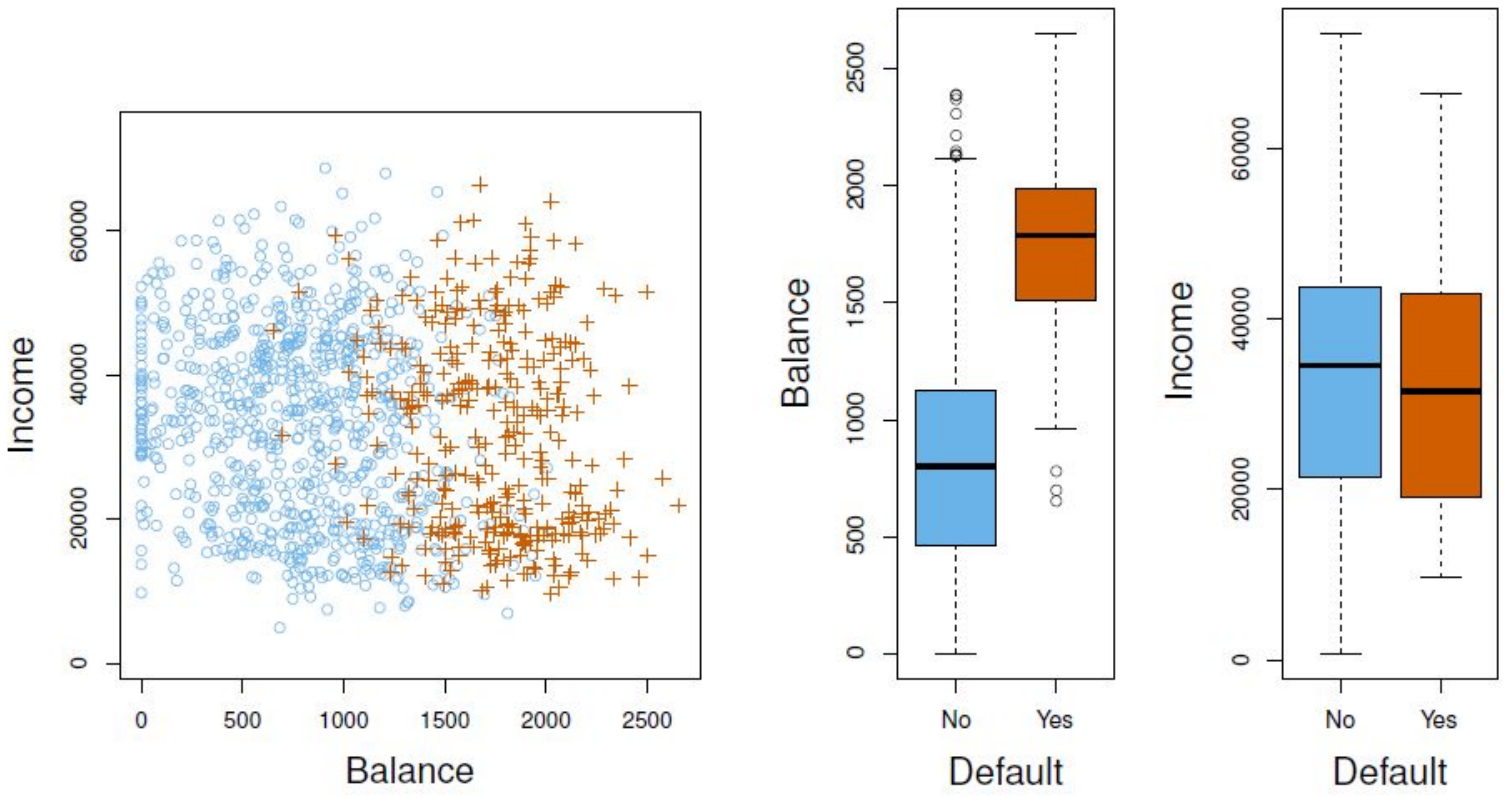
Regression in Context: Statistical Learning

Classification

- Qualitative variables take values in an unordered set $\mathcal{C}$, such as:
  `eye color`$\in$ {`brown`, `blue`, `green`}
  `email`$\in$ {`spam`, `ham`}.
- Given a feature vector X and a qualitative response Y taking values in the set $\mathcal{C}$, the classification task is to build a function $C(X)$ that takes as input the feature vector X and predicts its value for Y; i.e. $C(X) \in \mathcal{C}$.
- Often we are more interested in estimating the *probabilities* that X belongs to each category in $\mathcal{C}$.

  For example, it is more valuable to have an estimate of the probability that an insurance claim is fraudulent, than a classification fraudulent or not.

Example: Credit Card Default



Can we use Linear Regression?

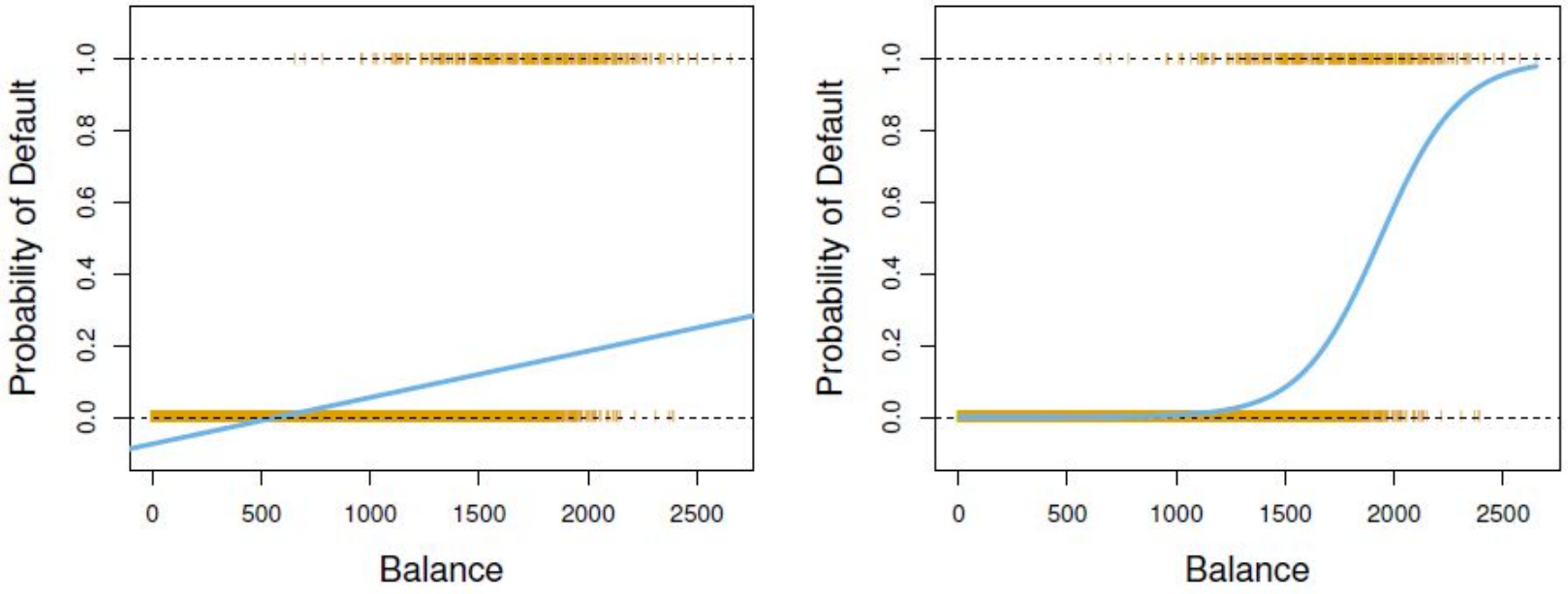
Suppose for the `Default` classification task that we code

$$Y = \begin{cases} 0 & \text{if } \texttt{No} \\ 1 & \text{if } \texttt{Yes}. \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as `Yes` if $\hat{Y} > 0.5$?

- In this case of a binary outcome, linear regression does a good job as a classifier, and is equivalent to *linear discriminant analysis* which we discuss later.
- Since in the population $E(Y|X = x) = \Pr(Y = 1|X = x)$, we might think that regression is perfect for this task.
- However, *linear* regression might produce probabilities less than zero or bigger than one. *Logistic regression* is more appropriate.

Linear versus Logistic Regression



The orange marks indicate the response Y, either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

Generalized linear models (GLM)

- This is a very general way of addressing many problems in regression]
- The resulting models are called generalized linear models (GLMs)
- Logistic regression is only one example
- All GLMs have the following three characteristics:
  - A probability distribution describing a generative model for the outcome variable
  - A linear model $\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$
  - A link function that relates the linear model to the parameter of the outcome distribution

Key Components of a Generaliized Linear Model (GLM)

Random Component

Distribution

Systematic Component

Linear Predictor

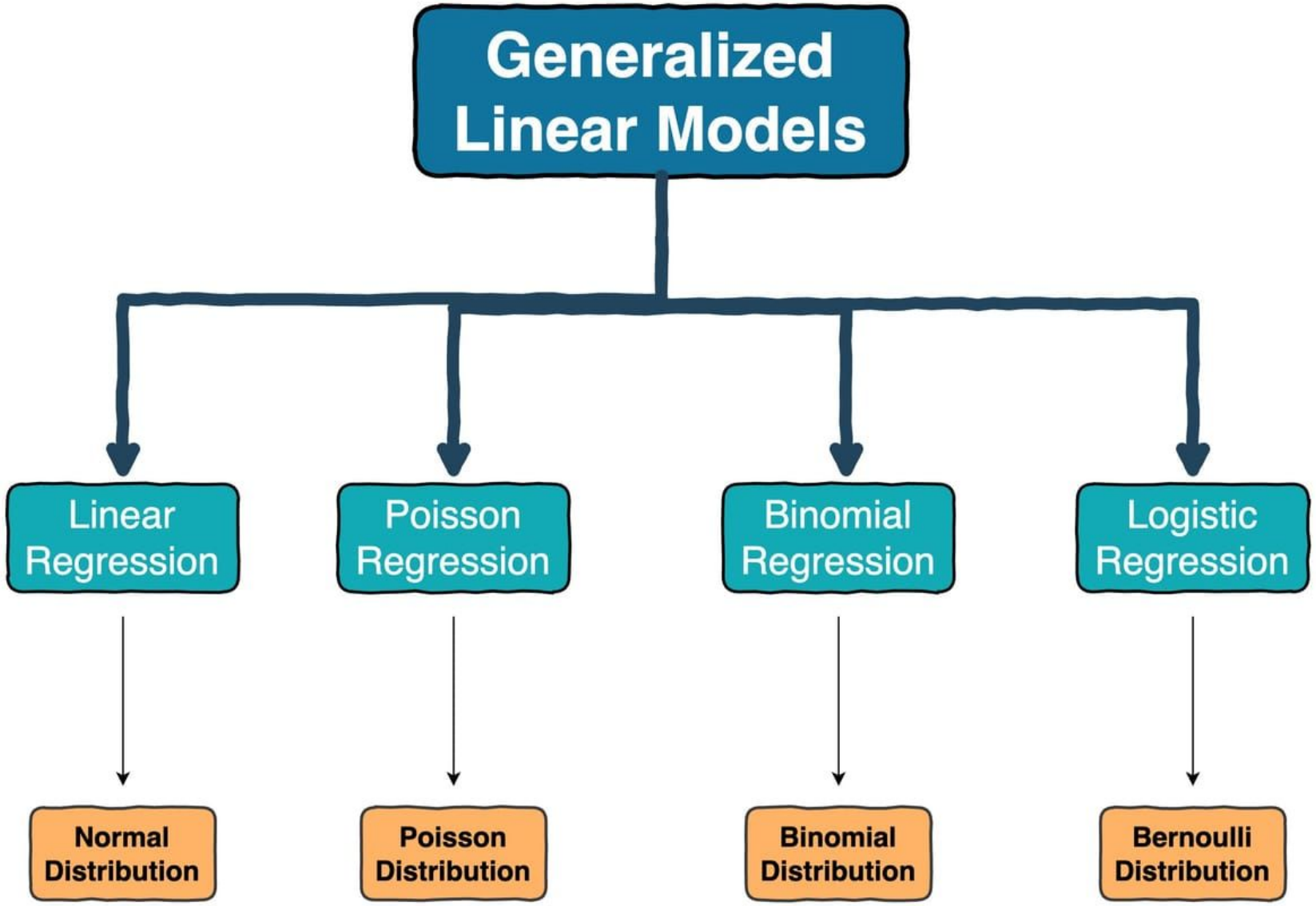
Link Function

$g(\mu) = \eta$

Common GLM Types

Linear Regression	Identity
Logistic Regression	Logit
Poisson Regression	Log
Gamma Regression	Inverse

Source: https://www.statology.org/a-beginners-guide-to-generalized-linear-models-glms/


Generalized Linear Models
Linear Regression
Poisson Regression
Binomial Regression
Logistic Regression
Normal Distribution
Poisson Distribution
Binomial Distribution
Bernoulli Distribution

GLM Type	Response Type	Distribution	Link Function
Linear Regression	Continuous	Normal	Identity
Logistic Regression	Binary	Binomial	Logit
Poisson Regression	Count	Poisson	Log
Gamma Regression	Skewed Positive	Gamma	Inverse

Source: https://www.statology.org/a-beginners-guide-to-generalized-linear-models-glms/

Framing the Linear Regression Classification Task

- We can treat each outcome (default or not) as successes and failures from separate Bernouilli trials

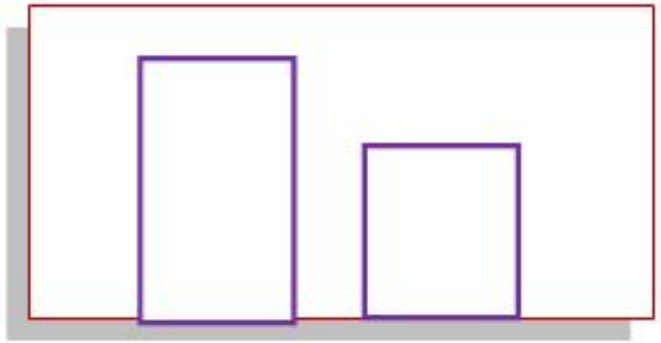
- Each Bernouilli trial can have a separate probability of success

$$y_i \sim Bern(p)$$

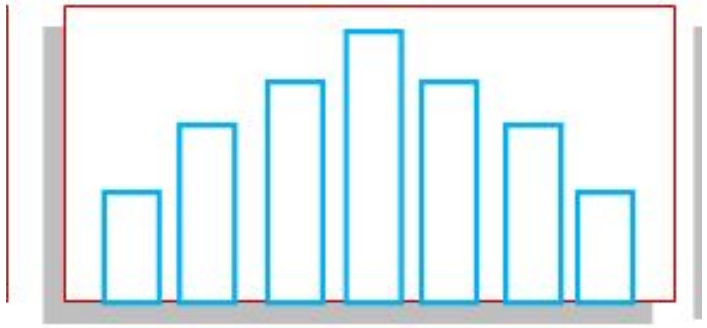
- We can then use the predictor variables to model *that probability of success*, p_i
- We can't just use a linear model for p_i (since p_i is a probability and must be between 0 and 1) but we can transform the linear model to have the appropriate range.

Key Differences

Aspect	Bernoulli Distribution	Binomial Distribution
Outcome	Single trial (one success or failure)	Multiple trials (count of successes)
Parameters	One parameter: p	Two parameters: n and p
Use Case	Binary outcome (single event)	Total successes in repeated trials

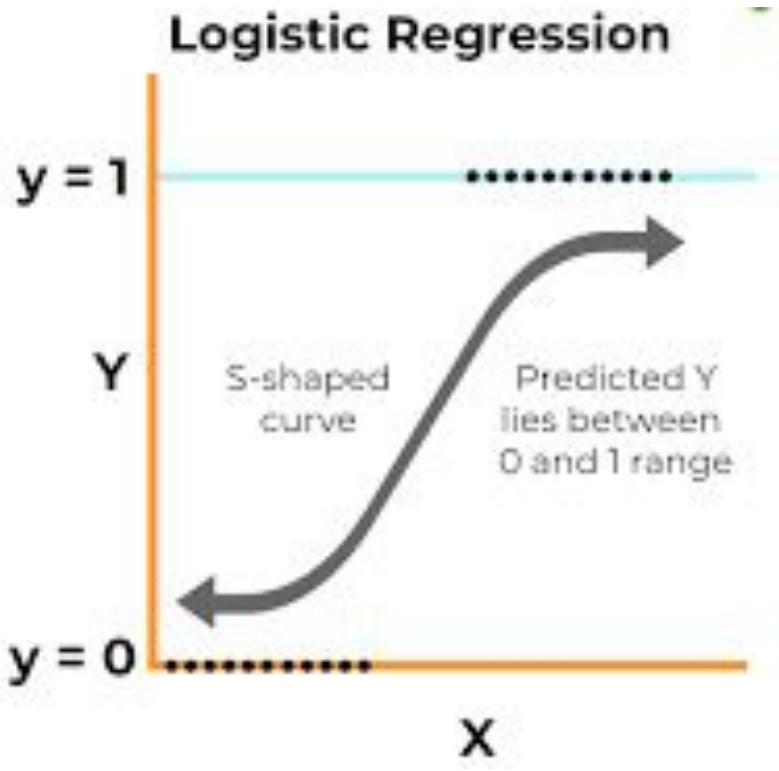


Bernoulli distribution



Binomial distribution

Quick Recap: Bernoulli Distribution and Binomial Distribution



a.k.a. **Log Odds**

or **Logit**

Intercept

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

Logistic Regression

- Logistic regression is a GLM used to model binary categorical outcome using numerical and categorical predictors
- The logistic regression is specified by
  - Bernouilli distribution
  - Linear function
  - Logit function (the linking function)

$$p_i = \frac{\exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i})}{1 + \exp(\beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i})}$$

Let's write $p(X) = \Pr(Y = 1|X)$ for short and consider using `balance` to predict `default`. Logistic regression uses the form

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1 + e^{\beta_0+\beta_1 X}}.$$

($e \approx 2.71828$ is a mathematical constant [Euler's number.])
It is easy to see that no matter what values β_0, β_1 or X take, $p(X)$ will have values between 0 and 1.

A bit of rearrangement gives

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

This monotone transformation is called the *log odds* or *logit* transformation of $p(X)$. (by log we mean *natural log*: ln.)

Some of the Logistic Regression Math

Maximum Likelihood

We use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

This *likelihood* gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.

Most statistical packages can fit linear logistic regression models by maximum likelihood. In `R` we use the `glm` function.

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-10.6513	0.3612	-29.5	< 0.0001
`balance`	0.0055	0.0002	24.9	< 0.0001

$$Odds = \frac{\#\text{for}}{\#\text{against}} \qquad P = \frac{\#\text{for}}{\#\text{for} + \#\text{against}} = \frac{\#\text{for}}{\#\text{total}}$$

$$\#\text{total} = \#\text{for} + \#\text{against}$$

$$odds = \frac{P}{1-P}$$

a.k.a. **Log Odds**

or **Logit**

Intercept

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X$$

$$\frac{P}{1-P} = e^{\beta_0 + \beta_1 X}$$

Intercept: the log of the odds of having the outcome, when X=0. The probability of having the outcome is $P = e^{\beta 0} / (1 + e^{\beta 0})$.

Coefficient: increasing the predictor by 1 unit multiplies the odds of having the outcome by e^{β}

Example: Interpreting Logistic Regression Coefficients

Model:

$$\text{logit}(p) = 0.5 + 0.13 * \text{study_hours} + 0.97 * \text{female}$$

p = P{Y=1} is the probability of passing a math exam. Increasing the study hours by 1 unit (1 hour) will result in a 0.13 increase in *logit(p)* or *log(p/1-p)*. If *log(p/1–p)* increases by 0.13, that means that *p/(1 — p)* will increase by *exp(0.13) = 1.14*. This is a 14% increase in the odds of passing the exam (*assuming that the variable female remains fixed*). Assuming study hours is fixed, the impact of being female is given by *exp(0.97) = 2.64*. The odds for passing the exam are 164% higher for women.

Source: https://towardsdatascience.com/a-simple-interpretation-of-logistic-regression-coefficients-e3a40a62e8cf

Making Predictions

What is our estimated probability of `default` for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 1000}}{1+e^{-10.6513+0.0055\times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 X}}{1+e^{\hat{\beta}_0+\hat{\beta}_1 X}} = \frac{e^{-10.6513+0.0055\times 2000}}{1+e^{-10.6513+0.0055\times 2000}} = 0.586$$

Lets do it again, using `student` as the predictor.

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-3.5041	0.0707	-49.55	< 0.0001
`student[Yes]`	0.4049	0.1150	3.52	0.0004

$$\widehat{\Pr}(\texttt{default=Yes}|\texttt{student=Yes}) = \frac{e^{-3.5041+0.4049\times 1}}{1+e^{-3.5041+0.4049\times 1}} = 0.0431,$$

$$\widehat{\Pr}(\texttt{default=Yes}|\texttt{student=No}) = \frac{e^{-3.5041+0.4049\times 0}}{1+e^{-3.5041+0.4049\times 0}} = 0.0292.$$

Logistic regression with several variables

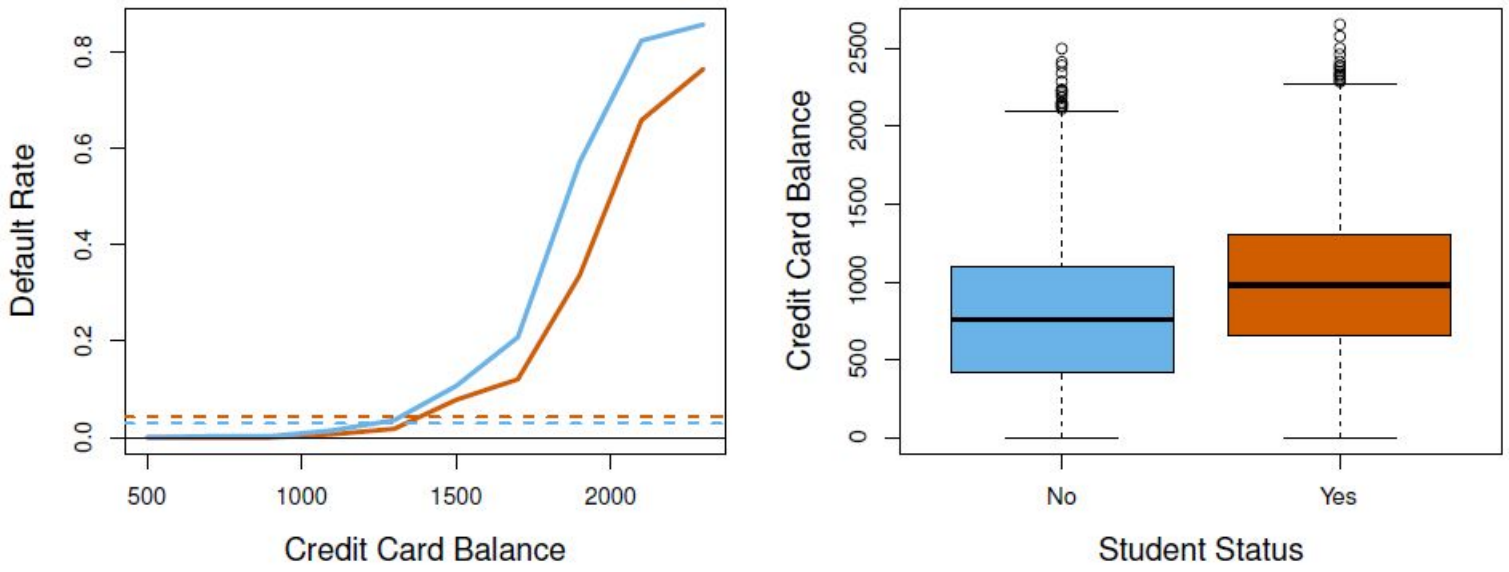
$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}$$

	Coefficient	Std. Error	Z-statistic	P-value
`Intercept`	-10.8690	0.4923	-22.08	< 0.0001
`balance`	0.0057	0.0002	24.74	< 0.0001
`income`	0.0030	0.0082	0.37	0.7115
`student[Yes]`	-0.6468	0.2362	-2.74	0.0062

Why is coefficient for `student` negative, while it was positive before?

Confounding



- Students tend to have higher balances than non-students, so their marginal default rate is higher than for non-students.
- But for each level of balance, students default less than non-students.
- Multiple logistic regression can tease this out.

Confusion Matrix

			Ground Truth Label	
	Total Observations (n)		*has disease*	*no disease*
			Condition Positive (CP)	Condition Negative (CN)
Predicted Label	*test positive*	Test Outcome Positive (TOP)	True Positive (TP)	False Positive (FP)
	test negative	Test Outcome Negative (TON)	False Negative (FN)	True Negative (TN)

Figure 1: Basic colour coded confusion matrix with marginal sums

Classification Metrics Formulas

	Predicted		
Actual	Positive	Negative	Row Totals
Positive	TP	FN	TotActPos
Negative	FP	TN	TotActNeg
Col Totals	TotPredPos	TotPredNeg	Total

$$\text{Recall} = \frac{\text{TP}}{\text{TotActPos}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TotAcNeg}}$$

$$\text{Error} = \frac{\text{\# Wrong}}{\text{Total}}$$

$$\text{Accuracy} = \frac{\text{\# Right}}{\text{Total}}$$

$$F = 2 * \frac{\text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TotPredPos}}$$

False positive and negative

	Email is spam	Email is not spam
Email labelled spam	True positive	False positive (Type 1 error)
Email labelled not spam	False negative (Type 2 error)	True negative

- False negative rate = P(Labelled not spam | Email spam) = FN / (TP + FN)
- False positive rate = P(Labelled spam | Email not spam) = FP / (FP + TN)

Sensitivity and specificity: Confusion Matrix (1)

	Email is spam	Email is not spam
Email labelled spam	True positive	False positive (Type 1 error)
Email labelled not spam	False negative (Type 2 error)	True negative

- Sensitivity = P(Labelled spam | Email spam) = TP / (TP + FN)
  - Sensitivity = 1 − False negative rate
- Specificity = P(Labelled not spam | Email not spam) = TN / (FP + TN)
  - Specificity = 1 − False positive rate

Sensitivity and specificity: Confusion Matrix (2)

Feature	`scikit-learn`	`statsmodels`
Focus	Prediction	Statistical inference
Adds intercept	Automatically	Must add manually (`sm.add_constant`)
Coefficient significance	Not provided	Includes p-values, standard errors
Scaling	Use `StandardScaler` or `Pipeline`	Must apply before fitting
Output	Accuracy and predictions	Detailed summary with diagnostics

Fitting Logistic Regression Models in Python